

A Markov Chain Model of Elite Table Tennis Competition

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ABSTRACT

The evaluation of the structure of sports performance is one of the important functions of diagnostics in competitive sport. Especially in game sports, it is important to obtain diagnostic information on competition because of the interactive process between the two teams or players. This interaction cannot be simulated or replicated in training or test situations. When it comes to table tennis, performance diagnostics offers many different techniques and methods to analyze a game. In this context, problems mostly occur in adequately modelling the game. Moreover, the approaches lack a scope of uniform performance criteria. According to a stochastic performance diagnostic concept (Markov Chain) for game sports, we developed four different state-transition-models to describe tactical behaviour in table tennis: (1) game action, (2) stroke position, (3) stroke direction and (4) stroke technique. Afterwards we formalized these models by means of finite Markov Chain (stochastic modelling) to determine the relevance of tactical behaviour to performance by simulation. In this study, 152 games of the top 50 ranked male players in the world were analyzed. The results suggest that in international-level table tennis, the diagonal game from backhand to backhand ("stroke position"), strokes into the long backhand zone ("stroke direction"), and especially the topspin stroke for shake-hand players in general ("stroke technique"), are the most important winning tactical strategies. In comparison with traditional methods used in elite table tennis the approach described here is valuable in quantifying and comparing the relevance of various tactical behaviours to performance.

Key words: Game Analysis, Performance Diagnostics, Stochastic Path Simulation, Tactical Behaviour in Sport

INTRODUCTION

Over many decades, research in game sports has focused on notational analysis by using game observation methods. The aims of notational analysts include optimizing feedback to the performer and coaches to improve performance [1]. Hence, the effectiveness of computerized notational analysis by means of systematic observation as quantitative and qualitative feedback on improving performance in game sports has been able to be demonstrated in various studies [2, 3].

From a performance-diagnostic point of view, the key tasks of systematic game observation are to structure the playing performance in all its complexity; i.e., to analyze the game structure and find determining factors, in this case tactics, to performance [4]. Literature research shows that a large number of methods have been developed – based on rapid technological developments in the area of computers, video etc. – to analyze the structure of game sports to identify performance indicators by game observation [1, 5]. A comprehensive review of the evolution of computerised notational analysis in racket sports is given by Hughes et al. [6].

Most of these methods use structure-oriented observation models, which enable the researcher to register isolated elementary actions in a match, but do not allow for the obtaining of data on the match process itself [7]. Hughes and Bartlett [1] point out, in this regard, that match classification indicators, such as ratio of shots per rally or ace rates, and technical or tactical indicators, such as winners and unforced errors, length of shots, distribution of winning shots and unforced errors, if used as indexing performance criteria in isolation, without reference to other normative data or opponents' data, are misleading.

This piece of game sports research thus views a game as a system in which the behaviour of the system is determined by, or dependent on, the interaction of the two playing parties. Regarding a sports game as a system or an interaction-process, several stochastic models were deployed within the framework of a sports game analysis. The first studies made use of probabilistic models (i.e., Markov chain) while concentrating on the calculation of probabilities to win the rally or to win the match [8-11]. Only in later research were technical and tactical aspects incorporated into the stochastic models. Here, the system of the “game” is described by categorical states strictly defined by individual techniques (e.g., stroke techniques in racquetball). Tactical behaviour then finds its expression in the transitions between these states. Models of the finite Markov Chains were successfully applied here, firstly to describe and predict playing profiles or patterns [12, 13], and secondly in order to determine the performance relevance of tactical behaviour [14-18].

Using a stochastic Markov model and the example of the game of squash, McGarry and Franks [12] were able to demonstrate, among other things, that player shot-response profiles significantly depend on the interaction between the two players; whereas, however, “the model is not, ..., generalizable to players beyond the scope of these player-player interactions”. Furthermore, in their study the authors were able to find that players used the same playing pattern against the same opponents (invariant behavioural responses) but different playing patterns when playing against different opponents (variant behavioural responses) [1, 19, 20].

In his attempt at performance diagnostics, Lames [14], in contrast to McGarry and Franks [12] does not use the stochastic model of the Markov Chain for the prognosis of future shot or response profiles, but for simulatively determining the types of tactical behaviour relevant to performance [21]. Given the fact that the tactical behaviour in game sports is dependent on the interaction between the players and can thus show great variability from game to game, it is important from the performance-diagnostic viewpoint to focus on the performance relevance of a tactical behaviour pattern, rather than on the frequency of its use.

In the contribution of Lames [14], comparable quantitative values for the performance relevance of selected tactical behaviour patterns are determined by the method of simulation, and for the purpose of prioritization can be ranked. This allows, for example, the investigation of whether we can determine invariants in performance relevant criteria; e.g., various performance-level groups, teams or player types, independent of inter- and intrapersonal variations in tactical behaviour.

In the final analysis, however, stochastic models are not suitable for investigating performance-relevant changes within a game (i.e., singular events which significantly influence the further course of a game). For this reason, system-dynamic concepts like relative phase, perturbations or antagonistic modelling are increasingly being applied in game sports research [22-26]. Here, the course of the game is conceived of as a dynamic process and the search for instabilities or phase transitions is made the focus of the game analysis. Rallies or game plays are isolated as units of analysis, while major emphasis is put on the identification of game-changing events. An application to tactics is currently only possible via the interpretation of data; and reference to additional information (e.g., video footage) remains indispensable here. The last of the aforementioned tasks of performance diagnostics lies in to what extent the system-dynamic analytical methods and procedures named above can contribute to structuring game performance, and in what form interpersonal comparisons can be carried out.

Proceeding from a performance-diagnostic aim, the following contribution applies the approach of Lames [14, 21] to the sport of table tennis. The investigation revolved around the question of the existence of differences in game structure; i.e., in tactical behaviour or performance-relevant criteria among world-class table-tennis players in general and dependent on bat-holding technique (shake-hand or penholder). In the study, four varying observation models were used to describe and analyse tactical behaviour.

METHOD

DESCRIPTION OF TABLE TENNIS

Table tennis is a competitive contest between two (single) or at most four (double) opponents. A game consists of several sets. In 2000 and 2001, international table tennis introduced rule changes which were expected to bring with them far-reaching changes to game structure. While ball-diameter was 38mm until 2000, since then a 40mm ball has been in use. Then, in 2001 a new scoring system became effective in which a set is completed when a player has reached eleven winning points (previously it had been 21).

Ball exchanges in table tennis usually last 2-4sec [27], while on average the number of strokes executed in singles rallies amounts to 5.13 (± 4.25) for women's, 4.32 (± 3.09) for men's and in doubles rallies 3.70 (± 2.20) for women's and 3.46 (± 2.25) for men's [28].

DATA COLLECTION - OBSERVATION MODELS

As a first step, it was necessary to develop a suitable system of observation appropriate to the goals. The course of the match was structured according to Lames [29] (Figure 1)¹.

To describe a table tennis match, the analysis unit is to be defined as the rally and the observation unit as the stroke (Figure 1). Hence, the table tennis match is described as a system that gradually moves through states, so that the tactical behaviour of each player is expressed in the transition between two subsequent states. So the term "*tactical behaviour pattern*", in the language of our model, is defined as the transition between two sequent states.

¹For further details about hierarchical structuring in game sports, see Hughes [30]

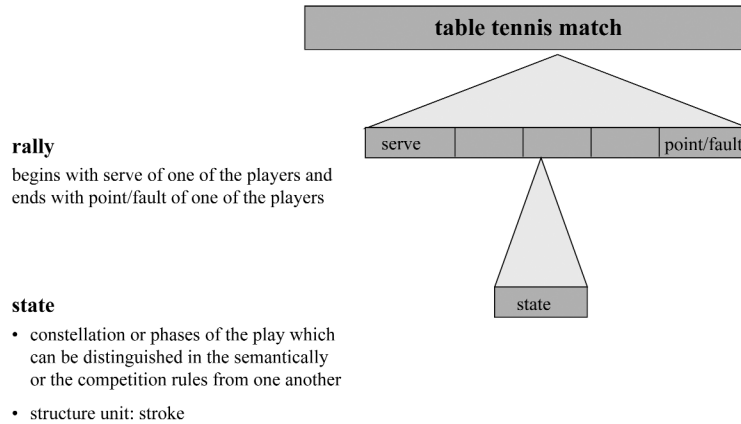


Figure 1. The Table Tennis Game Structured by Rallies and States

Furthermore, it is essential for the performance-diagnostic perspective to admit a performance criterion into the state system. The states of “point” and “fault” are defined as the criterion of performance, necessary for judging the performance relevance of a tactical behaviour pattern.

In our research, tactical behaviour in table tennis matches was examined by means of four different game observation models: (1) game action, (2) stroke position, (3) stroke direction and (4) stroke technique (Table 1). It is common for all four state systems to start off with a serve and end with a “point” for player A or player B.

The reason for using four models is that the many different tactical dimensions of the table tennis match, which for instance are expressed in the stroke position, stroke direction or technique, can hardly be tested with only one model. Consequently, the different results, reached by the four observation models should highlight the tactical behaviour underlying various aspects, which thereafter can be pieced together to form an overall performance-diagnostic picture.

The objectivity of the four observation models was confirmed by the agreement of two independent observers by using Cohen’s kappa statistics (Inter-Rater-Agreement) [31].

STOCHASTIC MODEL

By using a state-transition model, the course of the match can be described as the transition from one state to another. This process can be formalized as a stochastic model, in our case by using the Markov Chain theory.

A sports game like table tennis consisting of a sequence of trials (in this case strokes) with outcomes X_n ($n = 1, 2, \dots$), having probabilities satisfying the equation

$$P(X_n | X_1 \cap X_2 \cap \dots \cap X_{n-1}) = P(X_n | X_{n-1}) \quad (1)$$

is called a Markov chain. Thus, for such a chain, where the outcome of the current trial is known, the outcome of the next trial depends only on the current trial’s outcome and not on any additional information which may be available on the outcome of previous trials.

Table 1. State Models Used For Game Analysis in Table Tennis

Game action	Stroke position	Stroke direction	Stroke technique
Serve (S)	Serve (S)	Serve (S)	Serve (S)
Receive (R)	Forehand (FH)	Short Forehand (SF)	Topspin (T)
Neutral ¹ (N)	Backhand (BH)	Long Forehand (LF)	Drive (D)
Offence ² (O)	Pivot ⁵ (P)	Short Backhand (SB)	Smash (S)
Defence ³ (D)	Backhand turn ⁶ (Bt)	Long Backhand (LB)	Flip (F)
Control ⁴ (C)	Point (P)	Close to the Body (CB)	Chop (CH)
Point (P)		Net or Edge Ball (E/N)	Chopping short (CHs)
		Point (P)	Block (B)
			Cut (CU)
			Push (PU)
			Point (P)

¹Neutral: If the opponent strikes with attack technique (Topspin, Drive, Smash, Flip) and the observed player responds with attack technique as well, this was defined as “Neutral” from the observed player’s perspective.

²Offence: If the opponent strikes with control technique or defence technique (Chop, Chopping short, Push, Block, Cut) and the observed player responds with attack technique, this was defined as “Offence” from the observed player’s perspective.

³Defence: If the opponent strikes with attack technique and the observed player responds with defence technique, this was defined as “Defence” from the observed player’s perspective.

⁴Control: If the opponent strikes with control technique or defence technique and the observed player responds with control technique or defence technique as well, this was defined as “Control” from the observed player’s perspective.

⁵Pivot: If the observed player strikes in backhand position with forehand.

⁶Backhand turn: If the observed player strikes in forehand position with backhand.

Taking X_n to be the result of the n -th stroke of a match, X_n takes the value $A = \{a_n\}$ (for game action), $P = \{p_n\}$ (for stroke position), $D = \{d_n\}$ (stroke direction) and $T = \{t_n\}$ (stroke technique). In the terminology of Markov Chain theory $\{A, P, D, T\}$ is the state space associated with the sequence of trials. Each model reproduces the specific states that the original process went through.

The transitions from one state of the match to another can be described by its probabilities (as a percentage) and represents a tactical behaviour pattern of the players (see above). The transition probabilities describe the table tennis match as a system that gradually moves through discrete states (in our models “strokes”). This process can be understood as a Markov Chain under the acceptance of the following two properties: (1) Markov-property - the probability for the next state in time depending only on the current state; and (2) Chain-property - the transition probability from one state to another is independent of the chronological position of this state-transition in the match.

Following on from McGarry [32], Figure 2 illustrates how the sequence of strokes; i.e., the game process in a table tennis rally, can be analyzed by using a stochastic model (Markov).

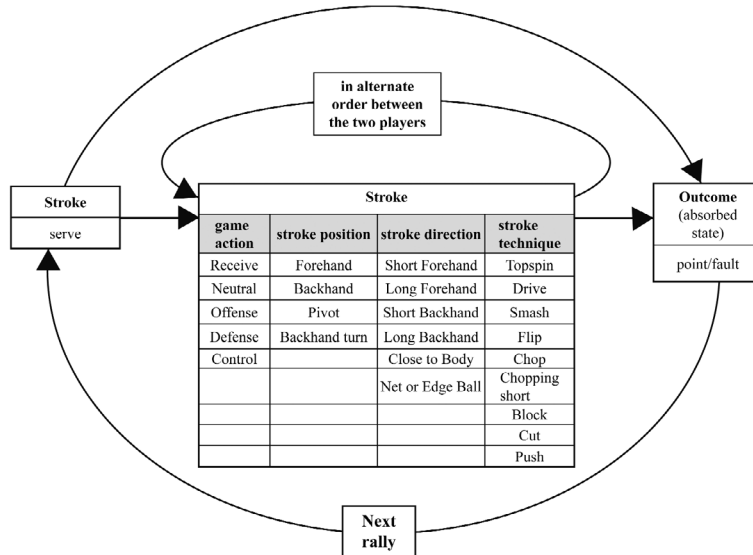


Figure 2. Stochastic Model (Markov) for the Sequence of Strokes and Outcomes Produced in a Table Tennis Rally and, by Extension, for all the Rallies in a Table Tennis Match Including the Used Four State Models (Modified from McGarry [22])

In any rally, the system begins in an initial state “serve” before passing to the next state, in our case “stroke”. Depending on the observation model (see Table 1) “stroke” can describe varying game situations. By defining these situations for both players, the alternating sequence of strokes in table tennis allows the interaction between the two opponents to be illustrated or modelled.

In the next step, the transition probabilities between the states can be transformed into a two dimensional transition matrix. Each element of this matrix has the property $p_{ij}(n) \geq 0$ and the line sum is equal to 1. In this way, the transition matrix provides a true picture of the interactive game process and provides an insight into the tactical behaviour of each of the two opponents. On the basis of the transition matrix, a tactical behaviour pattern can be defined as the transition probability between two states (Figure 3).

Figure 3 represents the transition matrix of the model “stroke direction” (see Table 1) using as an example the game between Jan-Ove Waldner (SWE) and Timo Boll (GER) at the 2002 European Championships.

The transition matrix reveals the following tactical characteristics:

- The stroke direction “Short Forehand” (SF) is the main aim of both players’ serve.
- At the transitions “Long Forehand” (LF) and “Long Backhand” (LB) leading to the opponent’s point, Waldner’s percentage of errors (15.63 and 28.33) is significantly higher than that of Boll (7.50 and 16.98). Both players show more errors in “LB” than in “LF”.
- Both players frequently play from “LF” to “LB”, especially Boll (75.00 %).
- Both players play to the opponent’s “LF” and “LB” from “LB” with approximately equal frequency.

	SF 1	SF 2	LF 1	LF 2	SB 1	SB 2	LB 1	LB 2	CB 1	CB 2	E/N 1	E/N 2	P 1	P 2
S_1		56.00		8.00		20.00		16.00						
S_2	68.00		12.00		16.00		4.00							
SF_1				31.58		5.26		47.37				5.26		10.53
SF_2			50.00				37.50						6.25	6.25
LF_1				28.13				50.00					6.25	15.63
LF_2			12.50				75.00						7.50	5.00
SB_1		50.00		25.00				25.00						
SB_2	33.33		16.67				33.33							16.67
LB_1				35.00				35.00					1.67	28.33
LB_2			38.30				39.62		7.55				16.98	7.55
CB_1				25.00				50.00						25.00
CB_2														
E/N_1														
E/N_2													100	

Figure 3. Example of a Transition-Matrix by Using the Observation System “Stroke Direction” – European Championship 2002; Team-Final Waldner (Player 1) Versus Boll (Player 2) (Abbreviations see Table 1)

On the basis of the transition matrix, it is possible to calculate some interesting characteristics of the progress of the game. In the theory of a Markov Chain, several kinds of states are distinguished. For the present case, absorbed states are important. The process stops in these states, it is absorbed and a new process starts. For the purpose of performance diagnostics, we define the state “Point” as an absorbed state. The Markov Chain theory now allows the calculation of some interesting characterizations of the process. For a performance diagnostic analysis, the transition probability to the absorbed states has preference. To quote Lames [14], this probability is called the *point probability (PP)*, and can be calculated for both players by multiplication of a start vector (distribution of the states) with the empirical transition matrix. In the present study, the starting vector for all matches of 50% for the two states “serve” of both players was assumed because of no other detailed knowledge of the state distribution. Multiplying the starting vector by the transition matrix results in a new vector, which in turn is multiplied by the transition matrix. This process is repeated by iteration until the probability of the non-absorbed conditions approaches zero (to 5 decimal points). In-so-doing, the probabilities for the two absorbing conditions of “point” are attained for both players. So, the PP describes the calculated probability of reaching the state “Point”; i.e., winning a rally (not a match), assuming the same number of serves.

SIMULATION TO QUANTIFY THE PERFORMANCE RELEVANCE OF TACTICAL BEHAVIOUR

Based on an observed (empirical) transition-matrix of a table tennis match, it is possible to calculate point probability (PP) based on a numerically manipulated (simulated) transition-matrix. In order to determine the performance relevance of a certain tactical behaviour pattern, the empirical transition probability between two states (representing the tactical behaviour pattern) was manipulated by a certain percentage in each observed match.

After this we once more calculated the point probability and defined the *performance relevance (δPP)* of a tactical behaviour pattern as the difference between the point probability (PP) as calculated by the original (observed) transition-matrix and the point probability as calculated by the manipulated transition-matrix.

In a first step, it is necessary to find a quantitative size of the manipulation of the investigated transition probability, so that the resulting values of the performance relevance

(δPP) remain in a comparable range.

Performance relevance is determined to a considerable extent by the quantitative size of the manipulation, meaning the comparability of this value requires the provision of a priority-catalogue of tactical behaviour patterns. The development of a process for the manipulation of tactical behaviour patterns, according to Lames [14] requires a specific modelling step. Such modelling is beset by the intrinsic problem of having to determine how scientifically plausibly the manipulations per tactical behaviour pattern can be realized. Because there is currently no theory on the difficulty of changing tactical behaviour, a mathematical function is proposed to serve as a model [14]. This is intended to provide the size of the manipulation dependent on the transitional probability of the respective behaviour pattern, as the sole independent variable with which the difficulty of such a change could be explained is the transition probability [14].

Since the available data does not provide scope for further specification of the model function, it is necessary, according to Lames [14], to justify the modelling from general principles on the difficulty of changing tactical behaviour patterns. Lames proceeds from the content-based assumption that the size of manipulation should be reciprocal to the difficulty of changing a tactical behaviour pattern. It follows that tactical behaviour patterns that are more difficult to change (as measured by the real transition probability), are represented by small manipulations, and that those of lesser changing difficulty are manipulated by greater changes in transition probability. From the original qualities, Lames [14] therefore extrapolates the following requirements for the model function:

1. Behaviour patterns in the mid-range of probability seem to be easier to change by a certain value than those at the edges. For the model-function, this means that the manipulation at the middle of the interval [0,1] should be greater than at the edges.
2. The difficulty of changing a tactical behaviour pattern rises towards the two limiting probabilities of 0 and 1, resulting in the degree of freedom declining in both directions. The model-function should thus be symmetrical about the point 0.5.
3. As the limiting probabilities of 0 and 1 can indeed be achieved in table tennis, balancing must be undertaken in the limit points of 0 and 1, because the corresponding behaviour patterns are intrinsically changeable. At the limit probabilities, the model-function assumes a manipulation value > 0 .

According to Lames [14] the function to deflect the transition probabilities used is

$$\delta TP(TP) = C + B \cdot 4 \cdot TP \cdot (1-TP) \quad (2)$$

Legend:

TP is the transition probability of the investigated tactical behaviour

δTP is the change of the transition probability of the investigated tactical behaviour

C is a constant, which describes the deflection in the border probabilities

B is a constant, which determines the maximum value of the relative portion of deflection

4 is a standardizing factor, so that the constant B is equal to the maximum value of deflection

The constant values applied in the present study were $C = 1$ and $B = 5$, which were also determined by Lames [14] and thoroughly tested by Pfeiffer [16]. The reason for the proposed values of the constants is that this specific combination leads only to minor correlations of the performance relevance (δPP) with the transition probability (TP) on the one hand, and the point probability (PP) on the other hand [14, 16]. With the chosen value combination, the transition probabilities were raised (manipulated) by 1 % in the minimum; i.e., at 0 % TP and by 6 % in the maximum, i.e. at 50 % TP (Figure 4).

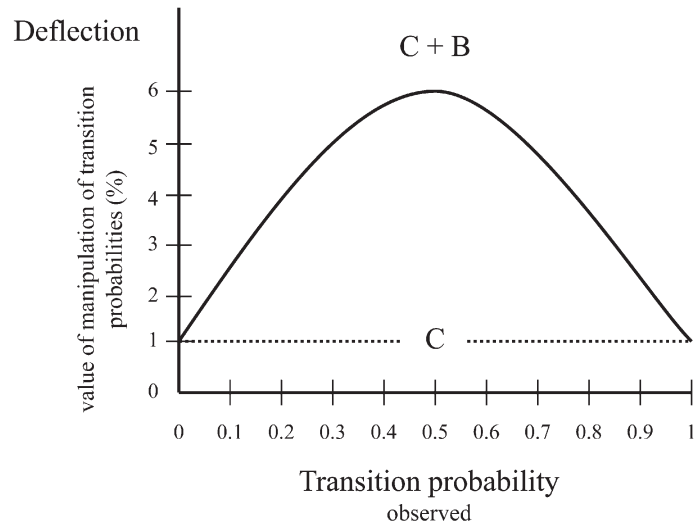


Figure 4. The Model Function: Amount of the Deflection in Relation to the Empirical/Observed Transition Probability

In previous studies, it has been possible to show that the effect of Parameter C on the extent of performance relevance is not dependent on the respective value of Parameter B. This means that although the numeric result for performance relevance is dependent on the choice of parameter, the relation between the results for all parameter combinations is essentially preserved [14].

In the next step, this procedure to calculate the relevance of performance requires establishing an algorithm that determines how the other, not manipulated transition probabilities have to be changed, so that the stochastic character of the matrix (sum of each row equals 1.00) is preserved. Therefore, an increase of one TP must be compensated for by a decrease of other transition probabilities in this row. In this study, each compensation procedure was based on a proportional compensation of all other transition probabilities, as no *a priori* information existed to justify undertaking a targeted compensation.

$$\delta TP_{yi} = -(TP_{yi}/(1 - TP_x)) \cdot \delta TP_x \quad (3)$$

Using the example of Figure 3, it can be demonstrated how the performance relevance (δPP) of a tactical behaviour pattern is calculated by a simulation on the basis of a standardized manipulation of the transition probability (Figure 5).

	SF_1	SF_2	LF_1	LF_2	SB_1	SB_2	LB_1	LB_2	CB_1	CB_2	E/N_1	E/N_2	P_1	P_2
S_1		56.00		8.00		20.00		16.00						
S_2	68.00		12.00		16.00		4.00							
SF_1				31.38		5.24		47.37				5.25		10.53
...				27.99		4.66		53.36				4.66		9.33
SB_2	33.33		16.67				33.33							16.67
...														
LB_2			38.30				39.62		7.55				16.98	7.55
...														

PP_1 (empirical matrix) = 54.4 Percent



PP_1 (manipulated matrix) = 55.0 Percent

$\delta PP_1 = 0.6$ Percent

Figure 5. Example of the Simulative Calculation of the Performance Relevance (dPP) of the Tactical Behavior of Player 1 “Strikes More Frequently from Short Forehand to Long Backhand of the Opponent” (Observation Model: Stroke Direction)

In the example of Figure 5, the increase of the transition probability between the states “SF_1” to “LB_2” by 5.99 % elevates the PP from 54.4 % to 55 %. According to the above-mentioned definition, the difference of 0.6 % documents the performance relevance (δPP) of the tactical behaviour pattern of player 1 “striking more frequently from short forehand to long backhand of the opponent” (Figure 5).

Thus the performance relevance (δPP) of tactical behaviour pattern is quantified by the scale in which the point probability (PP) changes if the transition matrix has been modified in terms of the tactical behaviour. Hence, the performance relevance is not calculated by means of the level of coherence of cause variable and performance criterion. Instead, it can be read off directly as a measured value of the simulated point probability. This comprises interval-scaled values, so that a rank order of the performance relevance of tactical behaviour pattern can be determined.

PARTICIPANTS AND ANALYSIS STATISTICS

In this study, 152 matches of the top 50 international male players were assessed at different official competitions over a period of 3 years (1997 to 2000). All matches were recorded on video at official competitions, so all subjects were accustomed to being observed and no written consent from the subjects was necessary. It is widely held that tactical behaviour and its performance relevance depend on the particular racket grip. Hence, for the game analysis of tactical behaviour, the matches were divided into four types of bat holding techniques: (1) shake-hand vs. shake-hand player ($n = 78$), (2) shake-hand vs. penholder player ($n = 25$), (3) penholder vs. shake-hand player ($n = 28$) and (4) penholder vs. penholder player ($n = 21$). Theoretically, there are only three kinds of pairing in the competition: shake-hand vs shake-hand, shake-hand vs penholder and penholder vs. penholder; and we only need to collect

three groups of data for the three pairings. But, in our research, for the “shake-hand vs penholder” pairing, we collected two separate player groups: one from the shake-hand player perspective, and another from the penholder player perspective. We did so mainly because shake-hand and penholder are two different kinds of playing styles, and we felt it necessary to perform separate analyses to compare the tactical behavior patterns and their performance relevance, not only between the match types, but also between the two player types. The comparison between different matches and players of bat holding styles is important for coaches and players with regard to strategic preparation in the run-up to subsequent matches.

The differences between the four types of matches were analyzed by one-way ANOVA and a post-hoc Scheffé test was applied to determine the origins of any significant differences. If the data of the various match groups did not show homogeneity of variance, the Kruskal-Wallis test (χ^2 statistic) was used. In all tests, $p < 0.05$ was accepted as being statistically significant.

RESULTS

OBJECTIVITY OF THE GAME OBSERVATION MODELS AND MODEL FIT

The instrumental consistency of the four observation models (objectivity) was examined by the inter-rater consistency of two observers (inter-observer agreement) quantified in Cohen's Kappa. Taking the different game types into account, 10 sets (1920 strokes or states) were selected from the examined games for this purpose. The Cohen's Kappa values (κ) of the observation systems were found to be $\kappa = .909$ for the “stroke technique”, $\kappa = .945$ for the “stroke direction”, $\kappa = .946$ for the “game action” and $\kappa = .969$ for the “stroke position” which according to Robson [31] represents an “excellent” classification (> 0.75).

With regard to the procedure chosen here for the determination of performance relevance in tactical behaviour patterns by means of simulation with the mathematical model of the Markov-Chain, it is of primary importance to ascertain whether the central calculated dimension of “point probability” (PP) for every game can show reference to the real success of both players. In order to ascertain model fit, it is standard practice to determine the relationship between empirical values with modelled values, because only the rank order of the two series is of importance with regard to the simulation results.

So the correlation coefficients between the observed and the mathematically modelled point probabilities (PP) for each observation systems were calculated to verify model validity [33]. The values varied between $r = .998$ and $r = .999$ ($p < 0.001$), which demonstrated the validity of the Markov model.

A static comparison of the probabilities of achieving individual conditions is of lesser importance to the aims of modelling described above. However, previous studies in game sports have shown that this is largely achieved to a satisfying degree [14, 16].

With regard to model validity, the fulfilling of the two basic pre-conditions for the use of a first order Markov-Chain (see chapter “Stochastic model”) have to be discussed: (1) each state of the table tennis rally only depends on the last former state and is independent from all other earlier states in the stochastic process, and (2) the transitions between the different states is independent of the position of these conditions in the sequence of events. As reported by Lames [14], both properties cannot be proven by a “hard” mathematical test, but must be established on the strength of content and logic. With the Markov-property, two problem clusters are addressed at the return of serve. “Memory Length”; i.e., for how long previous actions (here strokes) influence transition probabilities, as well as “Planning Depth”; i.e., how many actions are able to be planned in advance. Because of the short rallies and enormous time-pressure in men's singles table tennis (high ball velocity) (see

“Description of Table Tennis”, p. 207) only a slight level or depth for both concepts is to be assumed. In table tennis competitions, except for service, each stroke of the player is related to that of his opponent. That is to say, before each stroke, the player needs first to evaluate the stroke of his opponent (such as the placement, velocity, power, rotation, path, rhythm, etc.) and then act accordingly. This logically allows the assumption of an independence of transition probabilities from the previous state. Moreover, the high degree of correlation between real and mathematically calculated point probabilities (PP), point to the appropriateness of the model.

The second assumption merely describes transition probabilities per game comprising a mean value; i.e., the game as a whole is modelled as a “Super-Rally” [34]. In-so-doing, the statements on the tactical behaviour pattern of a player, or its performance relevance refer to the whole game, and thus represent an average behavioural value. Together with notational analysis, this is standard practice [5, 12, 19, 20, 32].

PERFORMANCE RELEVANCE OF TACTICAL BEHAVIOUR – DISTINGUISHED IN THE FOUR STATE SYSTEMS

The following presents only a selection of the results observed, as a complete account would go beyond the scope of this contribution. The results are presented for the four observation or state models (see Table 1), beginning with the reporting of peculiarities in the observed transition probabilities (TP). The most important findings on simulatively determined performance relevance (δPP) of tactical behaviour patterns in table tennis are presented. The low δPP values are the result of the deflection function selected (equation 2), which results in only a small degree of manipulation of the transition probabilities (TP); e.g., with a TP of 20 % by 4.2 %. The minimal deflection was chosen intentionally, as no intrinsically different behaviour was to be simulated. The manipulation serves to retrospectively determine the performance relevance of a certain component of tactical behaviour in an observed game. As a consequence of the complex transition matrices, such a minor manipulation of the empirical data only has a small effect on the absorbing states.

Game Action. In general, the results shows for all of the four match types that the direct point rate of each game action (transition to “Point”) was smaller than the corresponding fault rate (transition to “Point” of the opponent). The direct point rate of the “Offence” action was highest (11.8 %), followed by the “Neutral” action (6 %). The direct point rates of the other game actions were very low (0.5 % - 2.4 %). The fault rate of the “Defence” action was slightly over 50 %. The fault rate of the “Neutral” action amounted to 25 % and the fault rate of the “Offence” action was 17.3 %. This result shows that offence leads to a higher point rate and a comparatively lower fault rate.

The three highest measures of performance relevance (δPP) were detected in the tactical behaviour patterns: (1) both players strike with attack techniques (transition “Neutral” to “Neutral”), (2) one of the players forced the opponent to use a defence technique (transition “Neutral” to “Defence”) and (3) one player strikes against a control or defence technique with attack technique and the opponent answers that with attack technique (transition “Offence” to “Neutral”) (Figure 6). In the 152 matches, the mean of the performance relevance (δPP) of the tactical behaviour pattern expressed in the transition “Neutral” to “Neutral” amounted to 0.57 %, which was slightly lower than that of “Neutral” to “Defence”.

The performance relevance (δPP) of the transition “Neutral” to “Neutral” depended on the various game types: for example, δPP was higher in matches of shake-hand vs. shake-hand (SH-SH) players. The mean values of the other game types were significantly lower ($\chi^2_{3,151} = 24.64, p < 0.001$). When a shake-hand player competes with another shake-hand player, an

increase in the rate of the transition “Neutral” to “Neutral” has a greater influence on the match win. The average values of other game actions were below 0.50 %.

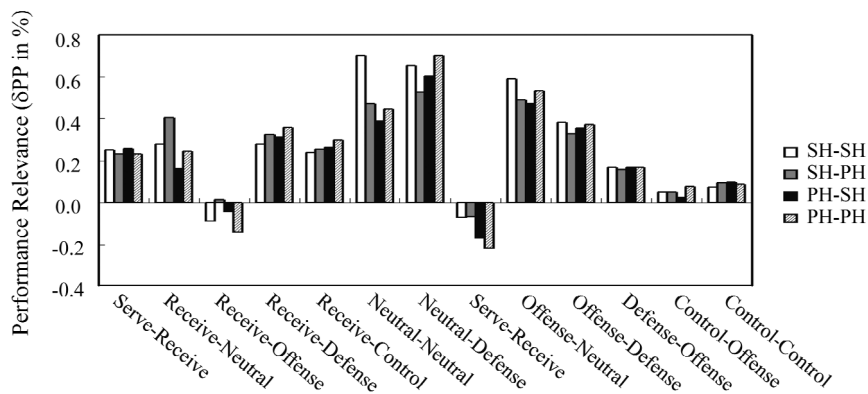


Figure 6. Performance Relevance (δPP) of Selected Tactical Behavior Pattern Collected by the Observation Model “Game Action”

Stroke Position. Here too, it can be shown that for all four match types the direct point rate of each stroke position (transition to “Point”) was smaller than the corresponding fault rate (transition to “Point” of the opponent). The direct point rate of the “Pivot” position was 9.3 %, which was much higher than those of the other stroke positions and therefore produced a better striking effect. The “Forehand” position followed with 5.84 %, whereas the direct point rate of the “Backhand” was relatively low (3.55 %). Fault rates were nearly the same in the “Forehand” and “Backhand” position (24.14 % vs. 25.19 %).

The tactical behaviour pattern “strike long line from the forehand side” (transition “Forehand” to “Backhand”) and “strike cross from the backhand side” (transition “Backhand” to “Backhand”) showed relatively high performance relevance (δPP) compared to other tactics (Figure 7).

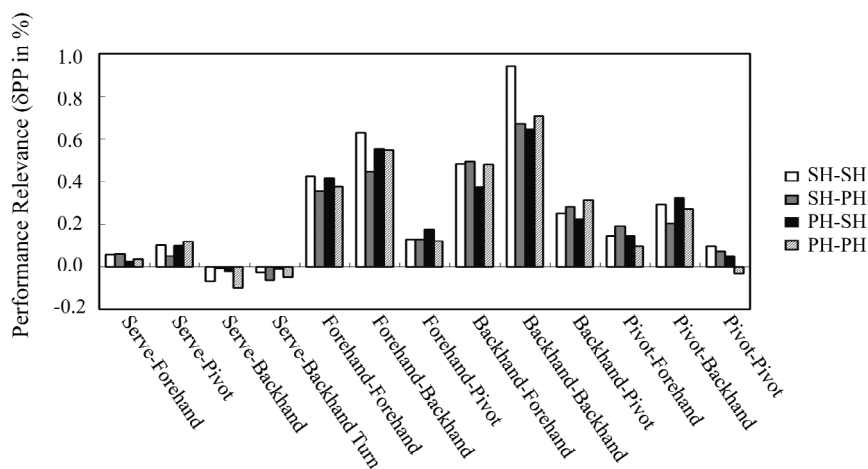


Figure 7. Performance Relevance (δPP) of Selected Tactical Behavior Pattern Collected by the Observation Model “Stroke Position”

The average δPP of the transitions “Forehand” to “Backhand” was 0.81 %, and post-hoc analysis revealed that in the play type shake-hand vs. shake-hand players (SH-SH) it was significantly higher ($p < 0.05$) than those of the other three play types ($= 8.83$, < 0.001).

Stroke Direction. For the transition probabilities, higher average error rates compared to direct point rates were observed for all stroke directions in all four match types. The direct point rates were nearly the same in the long forehand and backhand zone (7.43 % vs. 6.38 %), and the fault rate was a little higher in the long forehand zone compared to the long backhand zone. The fault rate was very high in the close-to-the-body zone (40.18 %), which indicates that “Close to the Body” is an effective stroke direction in table tennis.

The tactical behaviour pattern “answer a stroke into the long backhand also with a stroke into the long backhand” (transition “Long Backhand” to “Long Backhand”) showed very high performance relevance (δPP) (Figure 8).

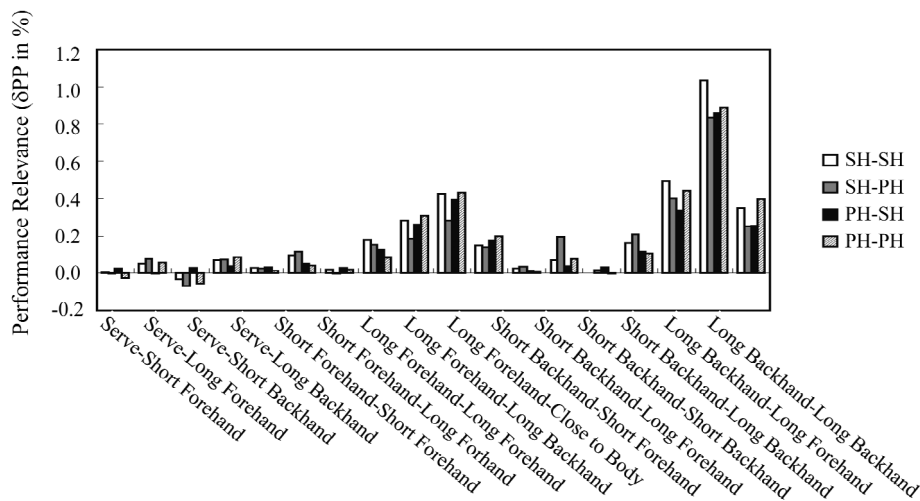


Figure 8. Performance Relevance (δPP) of Selected Tactical Behavior Pattern Collected by the Observation Model “Stroke Direction”

The average δPP amounted to 0.95 % in that tactic, so that an increase in the rate of this specific transition had a higher influence on the point probability than those of alternative transitions and respectively tactics for example “answer a stroke into the long backhand with a long forehand stroke” (transition “Long Backhand” to “Long Forehand”). Furthermore, the tactical behaviour pattern “response a stroke into the long forehand also with a long forehand stroke” (transition “Long Forehand” to “Long Forehand”) or “response a stroke into the long forehand with a long backhand stroke” (transition “Long Forehand” to “Long Backhand”) had lower performance relevance. Any differences between the four match types are, however, statistically insignificant.

Stroke Technique. As reported above for the other state models, conspicuously higher error rates than direct point rates also become evident for the stroke techniques. The direct point rate was the highest for the very effective technique “Smash” with 17.55 %, followed by the technique “Topspin” with 8.78 %. In addition, both techniques led to smaller fault rates than the other attack techniques “Drive” and “Flip”. The direct point and fault rates of “Drive” and “Flip” were similar (4.64 % vs. 24.10 %, and 4.17 % vs. 23.47 %). The direct

point rates of the two control techniques “Chop” and “Chopping short” were small, but the fault rate of “Chopping short” was only half of that of chop (12.2 %). The defence technique “Block” had by far the highest fault rate with 55.6 %.

Looking to the performance relevance (δPP), the tactical behaviour pattern “topspin technique against block technique” (transition “Topspin” to “Block”) was the greatest among all observed transitions between stroke techniques (Figure 9).

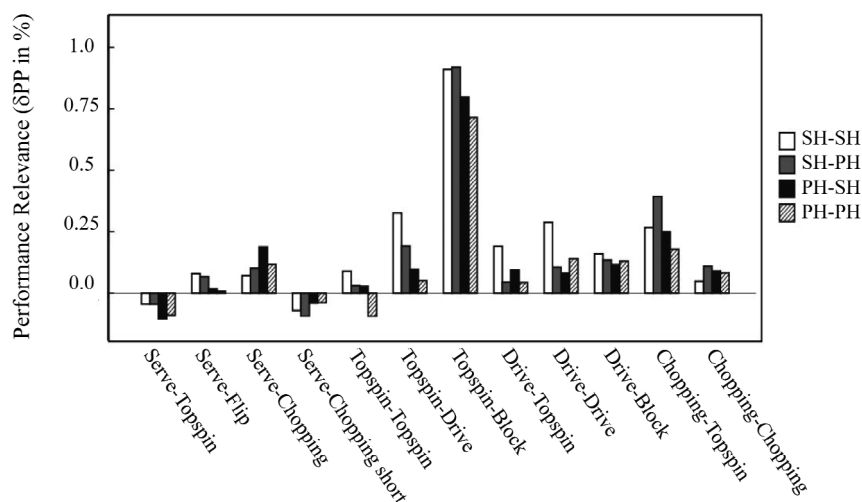


Figure 9. Performance Relevance (δPP) of Selected Tactical Behavior Pattern Collected by the Observation Model “Stroke Techniques”

The mean values of δPP for the transition “Topspin” to “Block” of the match type shake-hand vs. shake-hand player (SH-SH) and shake-hand vs. penholder player (SH-PH) were clearly higher than those of the play type penholder vs. shake-hand player (PH-SH) and penholder vs. penholder player (PH-PH) ($\chi^2_{3,151} = 17.09, p < 0.001$). Furthermore, the results for the shake-hand vs. shake-hand (SH-SH) group show significantly greater performance relevance for tactical behaviour pattern expressed in the transitions “Topspin to Drive” ($F_{3,151} = 8.57, < 0.000$), “Drive to Topspin” ($\chi^2_{3,151} = 28.33, p < 0.000$) and “Drive to Drive” ($\chi^2_{3,151} = 46.76, p < 0.001$). For the “Chopping to Topspin” transition, a significantly greater performance relevance results for the shake-hand player as opposed to the penholder player ($\chi^2_{3,151} = 4.87, p < 0.000$), which can also be confirmed statistically post-hoc (Scheffé). Conversely this means that when a penholder player competes against a shake-hand player, it is less performance-relevant to answer a chop with topspin than in the other match types.

DISCUSSION

Looking at the observed transition probabilities, it becomes apparent for all four state models – irrespective of a win or defeat – that the direct point rate (transition to “Point”) is smaller than the error rate (transition to “Point” of the opponent). This can be viewed as evidence that world-class players act with a greater willingness to take risks in any game situation. A fundamental reason for this high-risk approach may lie in the outstanding defensive performance of world-class players.

In general, “Neutral” and “Offence” are the two most important *game actions* in table

tennis. With regard to the transition probabilities (TP), the “Offence” action is still more important than the “Neutral” action, because of its better stroke effect (direct point rate).

Looking at the calculated values of the performance relevance (δPP), the “Neutral” action is more important than the “Offence” action, because the frequency of the “Neutral” action exceeds that of the “Offence” action. Therefore, an increase in the frequency of the “Offence” action and the effect of “Neutral” action are the main winning strategies in table tennis (Figure 6). From the high-performance relevance (δPP) for the transitions from the “Neutral” state, the action-recommendation, to respond to attack techniques with attack techniques, can be derived.

The results of the model *stroke position* show that from the backhand, the diagonal game (“Backhand to Backhand”) is more performance relevant than the parallel game (“Backhand to Forehand”), regardless of the match type or the bat holding style of the analysed player (Figure 7). This applies especially to shake-hand vs. shake-hand (SH-SH) games. In contrast, the performance relevance of parallel strokes from the forehand (“Forehand to Backhand”) is greater than for diagonal strokes (“Forehand to Forehand”). These results allow the action-recommendation both to play diagonally to the backhand from the backhand, and to place the ball in the backhand zone from the forehand, with greater frequency.

Moreover, the results show that the long backhand zone is a very important *stroke direction*. It constitutes a great advantage in comparison to the other stroke directions. An increase in the rate of strokes from the long backhand to the opponent’s long backhand position has a great impact on winning the rally for all groups (Figure 8). However, the performance relevance of this tactical behaviour pattern is significantly greater for games between two shake-hand players. As a consequence, world-class players are advised to play to the long backhand of the opponent from their own long backhand, instead of close-to-body or to the long forehand. From the long forehand, in contrast, they are advised to play close-to-body, as this tactical behaviour pattern shows greater performance relevance than the alternatives of “long forehand to long forehand” or “long forehand to long backhand”.

With regard to *stroke techniques*, the tactical behaviour pattern of “Topspin to Block” has an outstanding status independent of bat-holding style. For shake-hand players (SH-SH and SH-PH), this tactical behaviour pattern is significantly more performance relevant than for penholder players (Figure 9). All other tactical patterns relating to techniques seem to have little effect on the outcome of a rally. For the receiver, it is recommended wherever possible not to react to a topspin with a block, but with a further topspin (“Topspin to Topspin”) or with a drive (“Topspin to Drive”). Overall, most differences among the four match groups arise in stroke technique. For games between two shake-hand players, the performance relevance for the tactical behaviour patterns of “Topspin to Drive”, Drive to Topspin” and “Drive to Drive” is significantly greater.

CONCLUSION

In the present study, for performance diagnosis in table tennis, stochastic path simulation was used for the first time. In comparison with traditional methods, the main advantage of the approach as presented is that it delivers not only a statistical description of the analyzed tactical behaviour, but is also capable; of: a) analyzing the performance relevance of the various tactical behaviour patterns; and b) providing a prognosis on the probable effects of certain modifications to tactical behaviour patterns, based on mathematical simulation of the interactive game process. The performance diagnosis through stochastic path simulation by means of the Markov chain model is hence a worthwhile procedure in table tennis, allowing the calculation of the performance relevance of tactical behaviour patterns.

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